

2022-Ph-H1-Q17

Section: Particles and Waves

Topic: Nuclear Reactions

A nucleus emits an alpha particle. What changes?

Answer: D

(Mass number decreases by 4, atomic number decreases by 2)

Guidance for Students:

An alpha particle is a helium nucleus: ${}^4_2\text{He}$, so the original nucleus loses 4 nucleons and 2 protons.

Revision Tips:

- Alpha decay: lose 2 protons and 2 neutrons.
- Use nuclear equations to track changes.
- Alpha particles have high ionising power.